Assignment 9

221019225@link.cuhk.edu.cn

Please note that

- **Released date: December 4th, Wednesday.**
- **Due date: December 18th, Wednesday, by 11:59pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS9.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1. For each of the following, find a matrix B such that $B^2 = A$.

(a)

$$A = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$$

(b)

$$A = \begin{bmatrix} 9 & -5 & 3 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 2. Show that any 3×3 matrix of the form $\begin{bmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & b \end{bmatrix}$ is defective.

Question 3. Find all possible values of the scalar α that make the matrix

$$A = \begin{bmatrix} 3\alpha & 1 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{bmatrix}$$

defective or show that no such values exist.

Question 4. Let A be an $n \times n$ matrix with an eigenvalue λ of multiplicity n. Show that A is diagonalizable if and only if $A = \lambda I$.

Question 5. Let A be a diagonalizable $n \times n$ matrix. Prove that if B is any matrix that is similar to A, then B is diagonalizable.

Question 6. Show that if A and B are two $n \times n$ matrices with the same diagonalizing matrix X, then $AB = BA$.

Question 7. Find the matrix associated with each of the following quadratic forms:
(a) $3x^2 - 5xy + y^2$.
(b) $2x^2 + 3y^2 + z^2 + xy - 2xz + 3yz$.

Question 8. Show that if A is a symmetric positive definite matrix, then A is nonsingular and A^{-1} is also positive definite.

Question 9. Let A be an $n \times n$ matrix with positive real eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$. Let $\mathbf{x}_i$ be an eigenvector belonging to λ_i for each i, and let $\mathbf{x} = \alpha_1\mathbf{x}_1 + \cdots + \alpha_n\mathbf{x}_n$.
(a) Show that $A^m\mathbf{x} = \sum_{i=1}^{n} \alpha_i\lambda_i^m\mathbf{x}_i$.
(b) Show that if $\lambda_1 = 1$, then $\lim_{m\to\infty} A^m\mathbf{x} = \alpha_1\mathbf{x}_1$.

Question 10. Let A be a singular $n \times n$ matrix. Show that $A^\top A$ is positive semidefinite, but not positive definite.

Question 11. Let A be a symmetric positive definite matrix. Show that the diagonal elements of A must all be positive.

Question 12.

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$$A = \begin{bmatrix} 1 & 1 \\ 1 & a \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

For A and B, determine the condition of a such that A and B are positive define.

Question 13.Let

$$A = \begin{bmatrix} 0 & -2 & 1 \\ -2 & -3 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

Find an orthogonal matrix U that diagonalize A.